

Dr Shamendra Egodawela

Postdoctoral Research Fellow, UTS, Sydney

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Professional Summary

Academic and transdisciplinary researcher integrating AI, structural engineering, intelligent sensing, and stakeholder knowledge to solve real-world problems and translate research insights into teaching practice. Brings 6+ years of teaching experience across three universities, including curriculum and authentic assessment design, project-based learning, and direct supervision of 10+ research interns. Research record includes three Q1 journal articles and leadership of award-winning research in direct collaboration with industry stakeholders.

Education

2021 - 2025	RMIT University, Melbourne, Australia <i>Doctor of Philosophy in Civil Engineering</i>
2015 - 2020	University of Peradeniya, Sri Lanka <i>Bachelor of Science of Engineering in Electrical & Electronic Engineering</i>

Academic Experience

May 2025 - Present	University of Technology Sydney <i>Postdoctoral Research Fellow</i> - Designed and delivered postgraduate courses in Control of Mechatronic Systems, including course material, assessment activities, and project-based learning. - Developed an interactive wheeled-mobile-robot model predictive control demonstration linking mathematical concepts with simulation, optimisation, tracking error, and engineering judgement. - Designed postgraduate assessment methodologies that move beyond traditional coursework to include project demonstrations and structured vivas. This ensures students can confidently explain and defend their work, while providing a more reliable measure of understanding and authorship in the age of AI-assisted take-home assignments. - Lead Project SMaRTER with Amplitel, combining AI, structural engineering, and electronic sensing to support condition monitoring and asset-management decisions for telecommunications towers. - Supervise and coordinate 10+ research interns and multidisciplinary student teams from problem framing through system integration, experimental evaluation, and technical communication, managing internship delivery, evaluating progress, guiding and mentoring students throughout their research projects, and supporting industry visits and CRC-P field testing.
2023 - 2025	RMIT University <i>Casual Researcher & Course Assistant</i> - Delivered undergraduate lectures and laboratories in Mechanical, Manufacturing and Mechatronics Engineering, supporting hands-on, inquiry-led, and project-based learning. - Supervised multidisciplinary student teams developing an RFID-linked livestock sensing and automated weighing platform, including field installation and experimental evaluation. - Developed an augmented-reality-enabled physics-based digital twin integrating strain sensing, finite-element analysis, cloud communication, and mobile visualisation for teaching structural behaviour and delivering structural engineering courses.

University of Peradeniya
Teaching Instructor

- Delivered lectures, tutorials, and laboratory sessions in electrical machines, power systems, and robotics.
- Designed and assessed laboratory coursework involving electrical and embedded systems.
- Supervised undergraduate capstone projects in MATLAB, ANSYS, SOLIDWORKS, AutoCAD, and Arduino-based platforms.

Teaching & Supervision

- 6+ years of experience delivering undergraduate and postgraduate lectures, tutorials, laboratories, assessment activities, and project supervision to classes of 25+ students across UTS, RMIT University, and the University of Peradeniya.
- Align learning outcomes, interactive demonstrations, authentic assessment, and feedback so students connect theory with experimental evidence, engineering judgement, and implementation decisions.
- Use project-based learning, simulation, sensing, augmented reality, digital twins, project demonstrations, and structured vivas to help students examine complex systems, uncertainty, trade-offs, authorship, and real-world consequences.
- Helped organise UTS Tech Fest 2026, provided leadership in demonstrating and evaluating student projects, and served as a robotics showcase judge.
- Use Moodle and Canvas to manage course content, administer assessments, provide feedback, monitor student progress, and support student engagement; use Mentimeter and Kahoot! to facilitate interactive teaching sessions.

Research Program

Transdisciplinary research combining AI, engineering, sensing, and stakeholder knowledge to address complex problems in sustainable asset management and technology-rich learning.

- Develop transdisciplinary methods integrating machine learning, structural engineering, mechatronic sensing, digital twins, geospatial evidence, and operational knowledge to produce explainable and actionable outcomes.
- Publish in Q1 and leading international venues, including Nature, the *IEEE Sensors Journal*, and IEEE conferences, with outputs spanning AI, sensing, structural engineering, and infrastructure decision support.
- Build collaboration and networks across academia, industry, and education through research partnerships, student supervision, internship coordination, field evaluation, and technology-rich learning.

Research Output

Journal Publications

2025

1. **Egodawela, S.**, Gostar, A. K., Buddika, H. A. D. S., Harischandra, W. A. N. I., Dhammika, A. J., & Mahmoodian, M. Metal loss defect detection and depth estimation using multi-spectral image analysis of cooling excited steel specimen with corrosion. *Scientific Reports*, 15, 23894. doi:10.1038/s41598-025-88528-w [SJQR Q1]

2024

2. Mahmoodian, M., & **Egodawela, S.** Stochastic finite element methods for reliability analysis of corroded and fatigued steel elements. *SSRN*. doi:10.2139/ssrn.4975106
3. **Egodawela, S.**, Gostar, A. K., Buddika, H. A. D. S., Harischandra, W. A. N. I., Abeykoon, D., White, P. A., & Mahmoodian, M. Multi-spectral imaging and spectral unmixing for characterization of relative phase composition of steel corrosion products. *IEEE Sensors Journal*, 24, 37879–37893. doi:10.1109/JSEN.2024.3428472 [SJQR Q1]
4. **Egodawela, S.**, Gostar, A. K., Buddika, H. A. D. S., Dammika, A. J., Harischandra, W. A. N. I., Navaratnam, S., & Mahmoodian, M. A deep learning approach for surface crack classification and segmentation in unmanned aerial vehicle assisted infrastructure inspections. *Sensors*, 24(6), 1936. doi:10.3390/s24061936 [SJQR Q1]

Conference Publications

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| 2023 | 5. Egodawela, S. , Buddika, H. A. D. S., Harischandra, W. A. N. I., Mahmoodian, M., & Mathota, M. R. S. N. M. Conservation and management of ancient stupa using digital twins: A case study of UAV-based 3D photogrammetric digitization of Rankoth Wehera Stupa. In <i>International Conference on Sustainable Built Environment</i> , pp. 493–508. doi:10.1007/978-981-99-3471-3_35 |
| 2021 | 6. Perera, I., Senavirathna, S., Jayarathne, A., Egodawela, S. , Godaliyadda, R., Ekanayake, P., Wijayakulasooriya, J., Herath, V., & Sathyaprasad, S. Vehicle tracking based on an improved DeepSORT algorithm and the YOLOv4 framework. In <i>2021 10th International Conference on Information and Automation for Sustainability (ICIAfS)</i> , pp. 305–309. doi:10.1109/ICIAfS52090.2021.9606052 |
| 2020 | 7. Egodawela, S. , Herath, H. M. D. M. B., Willamuna, S. M. A. B., Ekanayake, M. P. B., Godaliyadda, G. M. R. I., Wijayakulasooriya, J. V., Herath, V. R., & Sathyaprasad, I. M. S. Vehicle detection and localization for autonomous traffic monitoring systems in unstructured crowded scenes. In <i>2020 IEEE 15th International Conference on Industrial and Information Systems (ICIIS)</i> , pp. 192–197. doi:10.1109/ICIIS51140.2020.9342663
8. Egodawela, S. , Herath, H. M. D. M. B., Ranaweera, R. D., & Wijayakulasooriya, J. V. Device to remotely track and locate the position of a child for safety. In <i>2020 IEEE Integrated STEM Education Conference (ISEC)</i> , pp. 1–7. doi:10.1109/ISEC49744.2020.9397849 |

Industry Engagement

- Work directly with Amplitel senior leadership and technical stakeholders to shape research directions, lead technical delivery and stakeholder engagement, and coordinate industry liaison, goals, risks, milestones, and outcomes.
- Participate in industry events such as CommsDay and industry information dissemination through Lunch and Learn sessions to maintain partner engagement and support project funding.

Technical Skills

Teaching / Curriculum	Course delivery, assessment, Moodle, Canvas, Mentimeter, Kahoot!
Programming	Python, MATLAB, C/C++, Arduino
AI / Data	Computer vision, NLP, predictive modelling, geospatial analysis
Engineering Tools	ANSYS, FEM, SOLIDWORKS, AutoCAD, embedded systems, sensors, PLCs
Digital Systems	Digital twins, robotics, cloud sensing, mobile AR, validation

Awards

CommsDay Edison Awards 2025 - Best Telco AI Initiative (Winner)

Project Lead, *Project SMaRTER* (Amplitel & UTS), an industry-scale AI and digital twin platform.

RMIT International Tuition Fee Scholarship 2021 (RRITFS)

Awarded a fully funded PhD scholarship through a competitive scholarship round.

Best Undergraduate Project in Electrical Engineering - Eng. E. W. Karunaratna Award 2019/2020

Institute of Engineers Sri Lanka (IESL)

Best Guest Tech Talk Award - NIBM Innovation Centre

Referees

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